

Problem

The amplitude of the voltage available in the 60-Hz, 120-V power outlet in your home is:

- (a) 110 V
- (b) 120 V
- (c) 170 V
- (d) 210 V

Step-by-step solution

Step 1 of 3

Consider the frequency of the input signal is **60 Hz**.

Consider the general equation for the sinusoidal input signal.

$$V_{in} = V_{rms} \sin(\omega t + \phi)$$

Consider the phase difference is 0.

$$V_{in} = V_{rms} \sin(\omega t)$$

Step 2 of 3

The maximum voltage available in the output is,

$$V_{max} = \sqrt{2} V_{rms} \sin(\omega t) \dots\dots (1)$$

The voltage at output is $\sqrt{2}$ times of the input voltage.

In the given options, option (a) and option (b) having voltages less than input signal voltages.

Therefore, option (a) and option (b) are wrong.

Step 3 of 3

Determine the maximum value of the output.

Replace **120** for V_{rms} , and **60 Hz** for f in equation 1.

$$\begin{aligned} V_{max} &= \sqrt{2} V_{rms} \sin(2\pi ft) \\ &= \sqrt{2} (120) \sin(2\pi (60)t) \\ &= 170 \sin(120\pi t) \text{ V} \end{aligned}$$

Therefore, the maximum value is **170 V**.

In the given options, option (d) is 210 V.

Therefore, option (d) is wrong.

Therefore, the option (c) is correct.